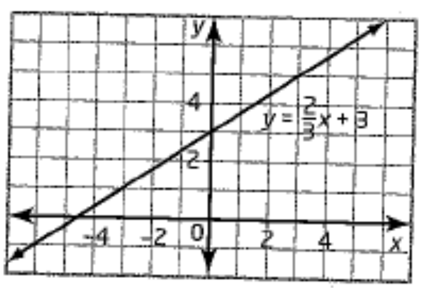
6.1 The Equation of a Line in Slope *y*-Intercept Form: $y = mx + b$, pages 296-307

1.

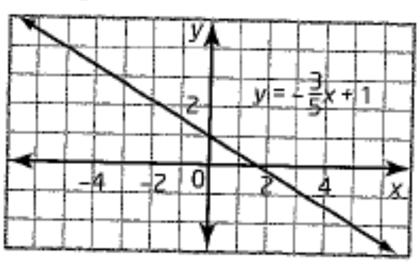
	Equation	Slope	*y*-intercept
a)	$y = 4x + 1$	4	1
b)	$y = \frac{2}{3}x + 3$	$\frac{2}{3}$	3
c)	$y = x - 2$	1	-2
d)	$y = -\frac{2}{3}x$	$-\frac{2}{3}$	0
e)	$y = 3$	0	3
f)	$y = -x - \frac{1}{2}$	-1	$-\frac{1}{2}$

2. **a)** slope 3; *y*-intercept -2
   b) slope -2; *y*-intercept 3
   c) slope $\frac{1}{4}$; *y*-intercept -2
   d) slope $-\frac{3}{4}$; *y*-intercept -2
3. **a)** $y = 3x - 2$ **b)** $y = -2x + 3$
   c) $y = \frac{1}{4}x - 2$ **d)** $y = -\frac{3}{4}x - 2$
4. **a)** $y = 2$; slope 0; *y*-intercept 2
   b) $x = -3$; slope undefined; no *y*-intercept
   c) $x = 4$; slope undefined; no *y*-intercept
   d) $y = 0$; slope 0; *y*-intercept 0
5. *x*-axis

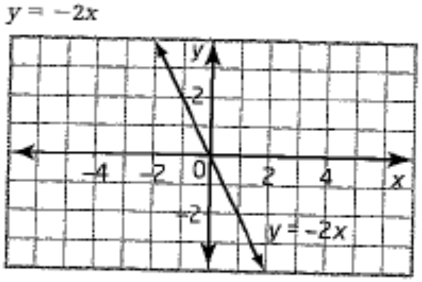
6. **a)** $y = \frac{2}{3}x + 3$



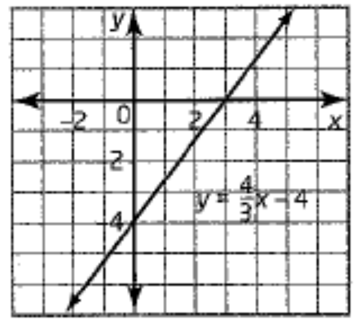
b) $y = -\frac{3}{5}x + 1$



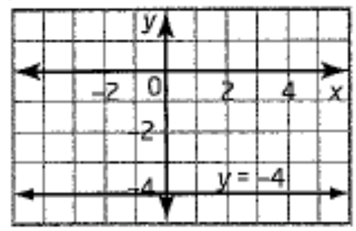
c) $y = -2x$



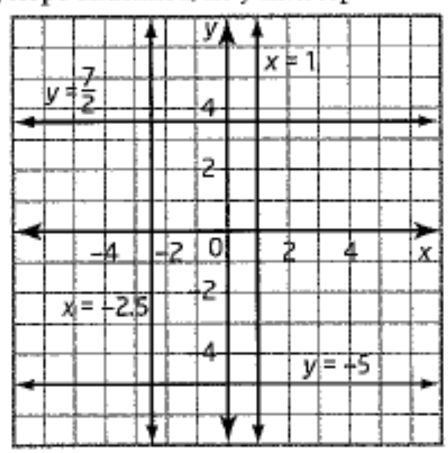
d) $y = \frac{4}{3}x - 4$



e) $y = -4$



7. **a)** slope 0; *y*-intercept -5
   b) slope undefined; no *y*-intercept
   c) slope 0; *y*-intercept $\frac{7}{2}$
   d) slope undefined; no *y*-intercept



8. **a)** The person was at an initial distance of 1 m from the sensor.
   b) The person was walking at a speed of 0.5 m/s.
   c) The person was walking away from the sensor. This is because on the graph, the person's distance from the sensor increases as time goes by.

10. **a)** slope 1; *t*-intercept 1.5; The slope represents Shannon's walking speed of 1 m/s away from the sensor. The *t*-intercept represents Shannon's initial distance of 1.5 m away from the sensor; $d = t + 1.5$.
    b) slope 3; *a*-intercept 0; The slope shows that the circumference of the trunk is three times its age. The *a*-intercept shows that when the tree began to grow from a seed, it had circumference zero. $C = 3a$.

12. Answers may vary